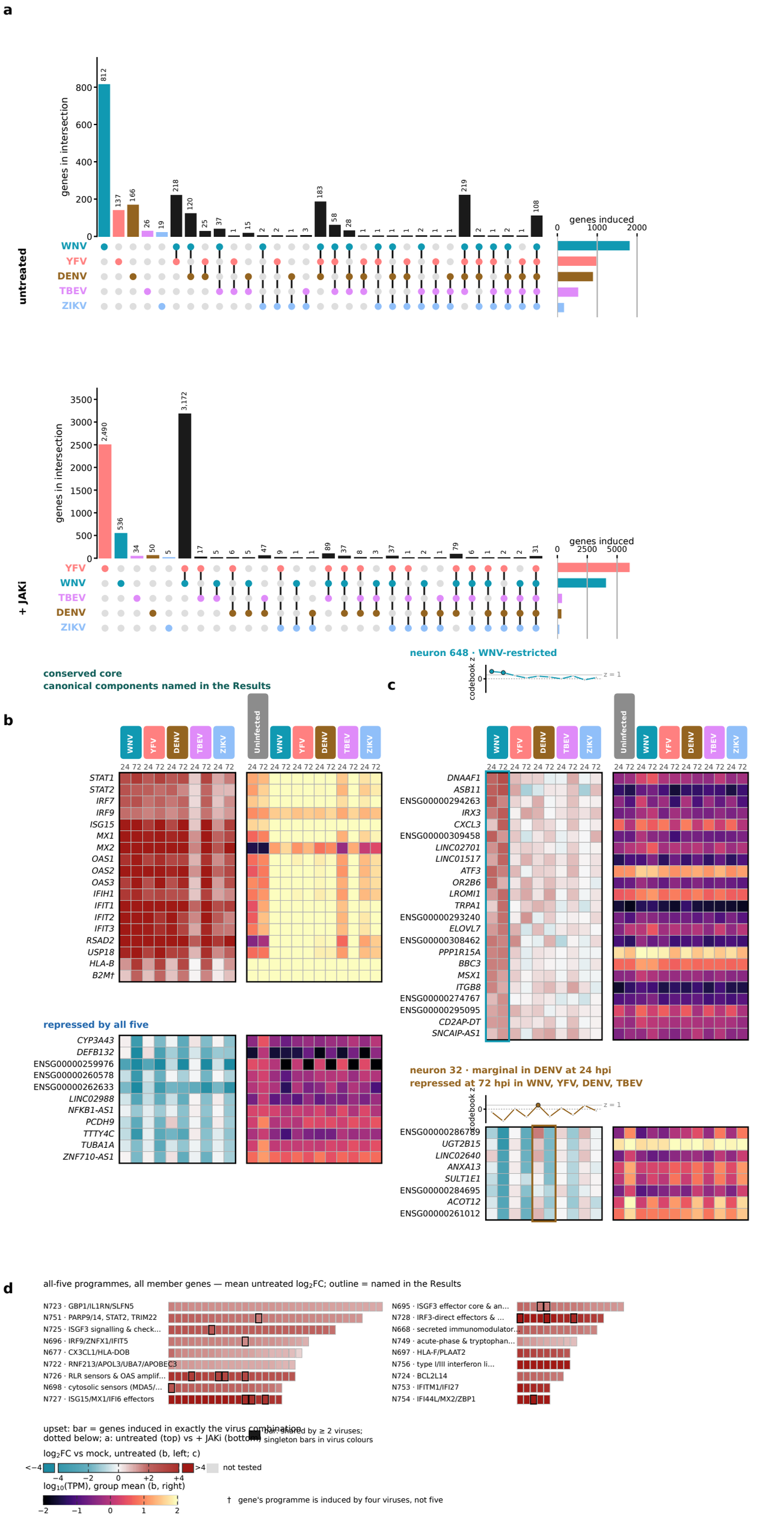

a
untreated
genes in intersection
812
137
166
26
19
218
120
25
37
1
15
2
2
1
3
183
58
28
1
1
1
1
2
1
1
219
2
1
2
1
108
WNV
YFV
DENV
TBEV
ZIKV
genes induced
0
1000
2000
+ JAKi
genes in intersection
2,490
536
34
50
5
3,172
17
5
6
5
47
9
1
1
89
37
8
3
37
1
2
1
79
6
1
2
2
31
YFV
WNV
TBEV
DENV
ZIKV
genes induced
0
2500
5000
b
conserved core
canonical components named in the Results
WNV
YFV
DENV
TBEV
ZIKV
Uninfected
24 72 24 72 24 72 24 72 24 72 24 72 24 72 24 72 24 72 24 72
STAT1
STAT2
IRF7
IRF9
ISG15
MX1
MX2
OAS1
OAS2
OAS3
IFIH1
IFIT1
IFIT2
IFIT3
RSAD2
USP18
HLA-B
B2M†
repressed by all five
CYP3A43
DEFB132
ENSG00000259976
ENSG00000260578
ENSG00000262633
LINC02988
NFKB1-AS1
PCDH9
TTTY4C
TUBA1A
ZNF710-AS1
c
neuron 648 · WNV-restricted
codebook z
z = 1
DNAAF1
ASB11
ENSG00000294263
IRX3
CXCL3
ENSG00000309458
LINC02701
LINC01517
ATF3
OR2B6
LROMI1
TRPA1
ENSG00000293240
ELOVL7
ENSG00000308462
PPP1R15A
BBC3
MSX1
ITGB8
ENSG00000274767
ENSG00000295095
CD2AP-DT
SNCAIP-AS1
neuron 32 · marginal in DENV at 24 hpi
repressed at 72 hpi in WNV, YFV, DENV, TBEV
ENSG00000286789
UGT2B15
LINC02640
ANXA13
SULT1E1
ENSG00000284695
ACOT12
ENSG00000261012
d
all-five programmes, all member genes — mean untreated log2FC; outline = named in the Results
N723 · GBP1/IL1RN/SLFN5
N751 · PARP9/14, STAT2, TRIM22
N725 · ISGF3 signalling & check…
N696 · IRF9/ZNFX1/IFIT5
N677 · CX3CL1/HLA-DOB
N722 · RNF213/APOL3/UBA7/APOBEC3
N726 · RLR sensors & OAS amplif…
N698 · cytosolic sensors (MDA5/…
N727 · ISG15/MX1/IFI6 effectors
N695 · ISGF3 effector core & an…
N728 · IRF3-direct effectors & …
N668 · secreted immunomodulator…
N749 · acute-phase & tryptophan…
N697 · HLA-F/PLAAT2
N756 · type I/III interferon li…
N724 · BCL2L14
N753 · IFITM1/IFI27
N754 · IFI44L/MX2/ZBP1
upset: bar = genes induced in exactly the virus combination; bar shaded by ≥ 2 viruses;
dotted below; a: untreated (top) vs + JAKi (bottom) singleton bars in virus colours
log2FC vs mock, untreated (b, left; c)
<−4
−4
−2
0
+2
+4
>4
not tested
log10(TPM), group mean (b, right)
−2
−1
0
1
2
† gene's programme is induced by four viruses, not five